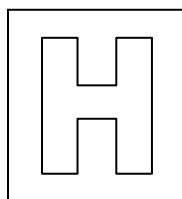


Candidate Name: _____

Class Adm No

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2015 Preliminary Examination 2

Pre-University 3

H2 Biology

9648/02

Paper 2 Core Paper

17 September 2015

2 hours

Additional Materials: Writing paper

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your Admission number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer any **one** question.

The use of an approved scientific calculator is expected, where appropriate. You will lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	
2	
3	
4	
5	
6	
7	
8	
Section B	
Total	

This question paper consists of 26 printed pages including 3 blank pages.

[Turn over

Section A

Answer **all** questions in this section.

1. The cell membrane is an important component for all cells. Its role is critical because its structural components provide the barrier that marks the boundaries of a cell. Fig. 1.1 shows part of the cell surface membrane of a liver cell.

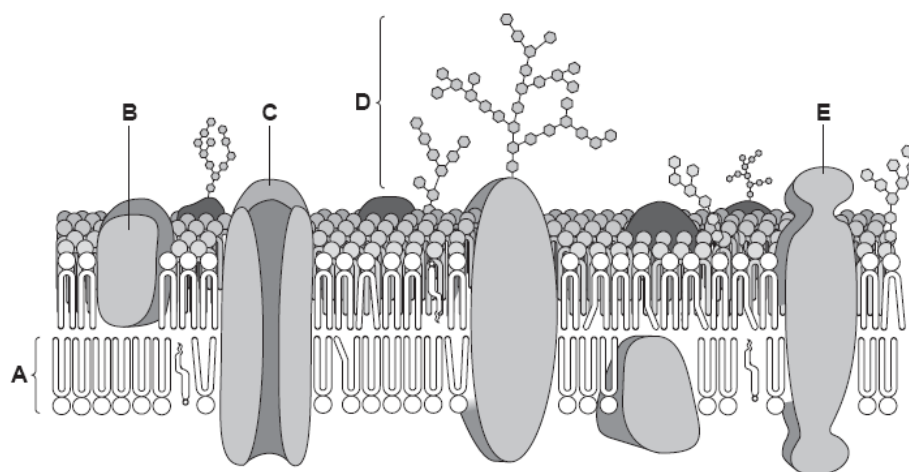


Fig. 1.1

- (a) Identify biomolecules **A** and **C**.

A

C

[1]

- (b) Describe the arrangement of biomolecules **A** in the membrane as seen in Fig. 1.1.

.....

[2]

- (c) Explain what determines the three-dimensional shape of biomolecule **C**.

.....

[3]

Biomolecule **D** is formed by joining many monosaccharides together.

- (d)** State the type of bond that joins monosaccharides together in biomolecule **D**

.....[1]

A large number of storage polysaccharides can be found throughout the liver cell.

- (e)** Explain how the structure of this storage polysaccharide makes it a good energy storage molecule.

.....

.....

.....[2]

[Total: 9]

2. Gout is a medical condition, which is usually characterised by recurrent attacks of acute inflammatory arthritis. It is caused by the build-up of uric acid crystals in joints. Uric acid is produced from xanthine in a reaction catalysed by the enzyme xanthine oxidase.

A medical scientist investigated the effect of increasing xanthine concentration on the rate of activity of mammalian xanthine oxidase. Ten test-tubes were set up with each containing 5 cm³ of different concentrations of xanthine solution. The test-tubes were placed in a water bath at 40 °C for ten minutes. A flask containing a xanthine oxidase solution was also put into the water bath. After ten minutes, 1 cm³ of the xanthine oxidase solution was added to each test-tube. The reaction mixtures were kept at 40 °C for a further ten minutes. After ten minutes, the temperature of the water bath was raised to boiling point. The amount of uric acid was measured and used to calculate the rate of enzyme activity.

Results of the experiment are shown in Fig. 2.1.

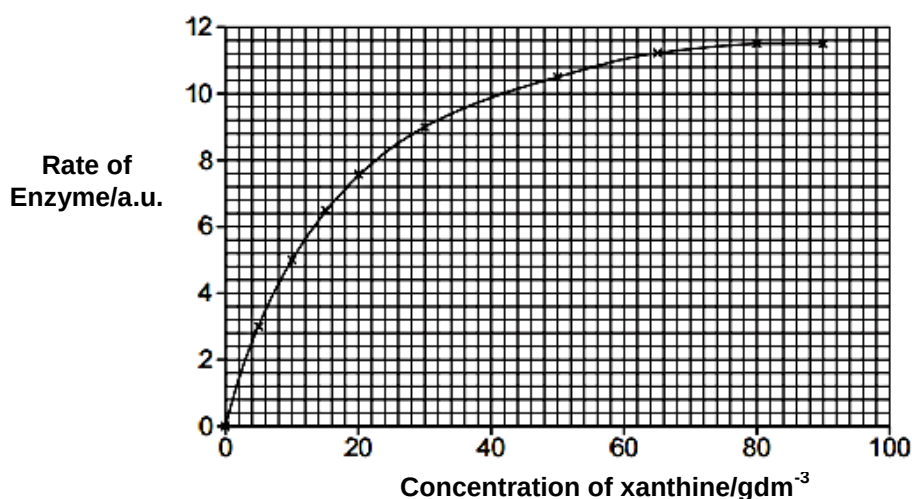


Fig. 2.1

- (a) Account for the shape of the graph shown in Fig. 2.1.

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.....[4]

Using similar experimental procedures, the student proceeded on to investigate the effect of temperature on the activity of mammalian xanthine oxidase and xanthine oxidase found in thermophilic bacteria living in hot springs.

- (b) Explain why xanthine oxidase in thermophilic bacteria has a higher optimum temperature than mammalian xanthine oxidase.

.....

[2]

Allopurinol is a drug used to treat gout. Fig. 2.2 shows the structures of xanthine and allopurinol.



Fig. 2.2

- (c) With reference to Fig. 2.2, suggest how allopurinol can be used to treat gout.

.....

[2]

[Total: 8]

3. Human T-cell Lymphotropic Virus (HTLV) was the first human retrovirus discovered. HTLV belongs to the Retroviridae family and predominantly affects T lymphocytes. HTLV has been implicated in several kinds of diseases including very aggressive adult T-cell lymphoma, which is the cancer of T-lymphocytes.

A molecule of messenger RNA (mRNA) was produced during the transcription of a HTLV gene in the host T lymphocytes. Part of the template sequence of DNA was ATGC. Fig. 3.1 shows the part of the molecule of messenger RNA corresponding to that sequence of four bases.

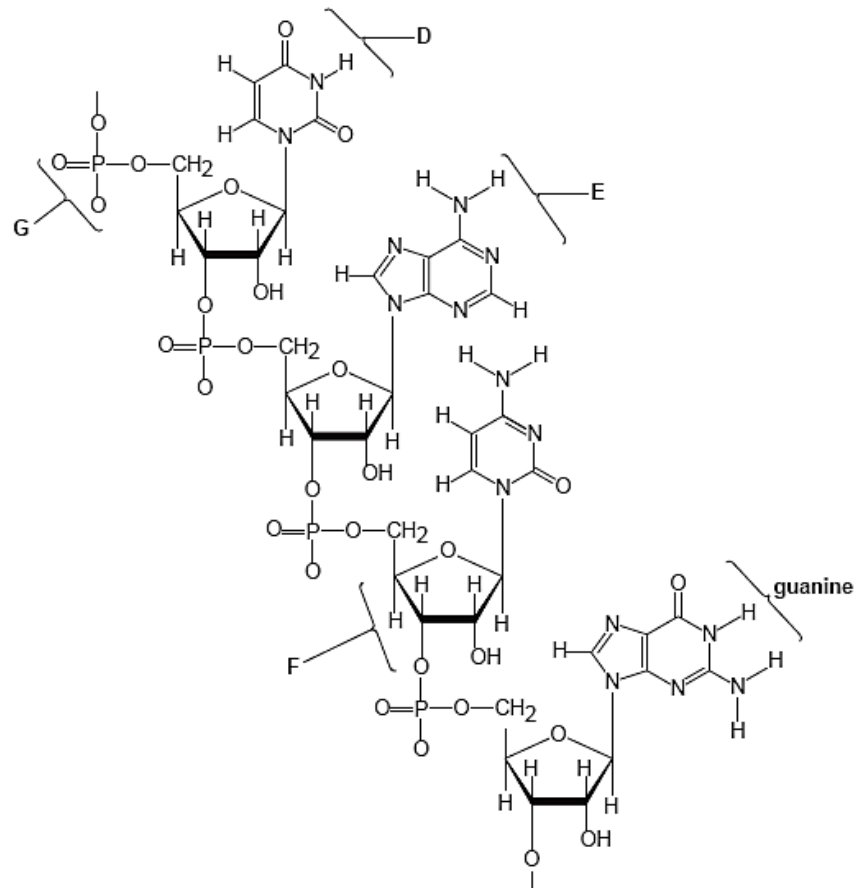


Fig. 3.1

- (a) Name the parts of the mRNA molecule shown in Fig. 3.1 labelled **D**, **E**, **F** and **G**.

D:

E:

F:

G:

[2]

- (b) Describe the role of mRNA after it leaves the host cell nucleus and enters the cytoplasm.

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..... [3]

Fig. 3.2 shows the reproductive cycle of HTLV, which closely resembles that of HIV. HTLV binds to the GLUT-1 receptors on the T lymphocytes. Its viral envelope contains two types viral glycoproteins (SU and TM) deriving from a common precursor.

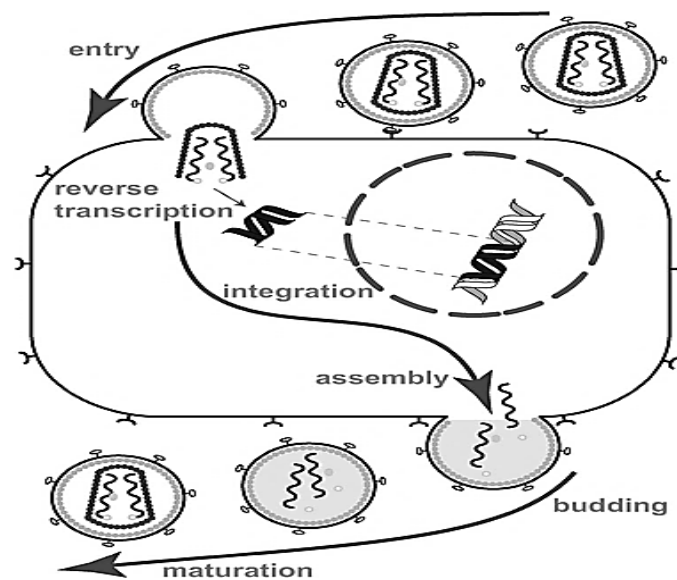


Fig. 3.2

- (c) With reference to Fig. 3.2, distinguish between the mode of entry of a T4 bacteriophage and HTLV.

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..... [2]

HIV spreads through body fluids and affects CD4⁺ T lymphocytes. Over time, HIV can destroy so many of these cells that the body cannot fight off infections and diseases.

(d) Describe the importance of reverse transcriptase in HIV reproduction.

.....

 [2]

Various drugs have been developed to treat HIV infections. Table 3.1 gives information about some of these drugs. HIV-1 surface glycoproteins are known to display antigenic variation due to antigenic drift. As a result, people, who receive drug treatment for HIV, have to take a mixture of drugs that act in different ways.

Table 3.1

drug	enzyme inhibited
zidovudine	reverse transcriptase
tenofovir	reverse transcriptase
efavirenz	reverse transcriptase
atazanavir	protease

(e) Using your knowledge of antigenic drift, explain the advantage of taking a mixture of the drugs shown in Table 3.1.

.....

 [2]

Adult T-cell leukaemia/lymphoma (ATL) is usually a highly aggressive non-Hodgkin's lymphoma. Several lines of evidence suggest that HTLV causes ATL. This evidence includes the frequent isolation of HTLV-1 from patients with this disease and the detection of HTLV-1 proviral genome in ATL leukemic cells.

- (f) Explain how the integration of the HTLV genome into T-lymphocytes' chromosome could lead to cancer.

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.....[2]

The median age at diagnosis for ATL is 56 years.

- (g) Using your knowledge on cancer development, suggest why ATL rarely appears in younger individuals.

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.....[1]

[Total: 14]

4. Penicillin is the first antibiotic to be discovered in 1928 by Alexander Fleming. Since then, antibiotics are being manufactured at an estimated scale of about 100,000 tonnes annually worldwide, and their use had a profound impact on the life of bacteria on earth. More strains of pathogens have become antibiotic resistant.

Penicillin resistance in *Streptococcus pneumoniae* involves point mutations in the gene coding for penicillin-binding proteins (PBPs). PBPs are the target enzymes of penicillin action. These mutations could result in reduced penicillin affinity.

- (a) Describe how a mutation in *PBP* gene may result in an enzyme with reduced penicillin activity.

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.....[2]

Some strains of bacteria have become resistant to practically all of the commonly available antibiotics. A notorious case is the methicillin-resistant *Staphylococcus aureus* (MRSA). It can act as a major source of hospital-acquired infections. An old antibiotic, vancomycin, was resurrected for treatment of MRSA infections. Molecular studies revealed that a plasmid-borne transposable element encodes enzymes and regulator proteins that confer resistance of MRSA to the antibiotic vancomycin.

- (b) Describe two ways by which vancomycin resistance may spread naturally amongst bacteria.

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.....[2]

In the absence of glucose, *Escherichia coli* can proliferate using the pentose sugar arabinose. The ability of *E. coli* to utilise the sugar arabinose is regulated via the arabinose operon, depicted in Fig. 4.1 below.

The structural genes (*araA*, *araB*, and *araD*) encode enzymes for the metabolism of arabinose. The *araC* gene encodes a gene regulatory protein that binds adjacent to the promoter of the arabinose operon.

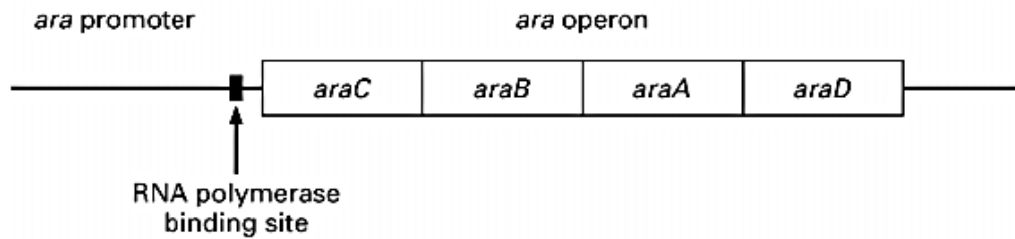


Fig. 4.1

(c) Distinguish between structural and regulatory genes.

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.....[2]

An experiment was conducted to investigate the role of araC regulatory protein in regulating arabinose operon. The results of the experiment are shown in Table 4.1.

Table 4.1

Genotype	araA RNA Levels	
	in the absence of arabinose	in the presence of arabinose
<i>araC</i> ⁺ (normal cells)	low	high
<i>araC</i> ⁻ (mutant cells)	low	low

- (d) With reference to the results in Table 4.1, explain whether araC regulatory protein functions as a repressor or activator in regulating the arabinose operon.

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.....[2]

[Total: 8]

5. The human β globin locus is located on chromosome 11. This chromosome contains not only the β globin gene but also ϵ , γ_G , γ_A and δ globin genes. Expression of all of these genes is controlled by single locus control region (LCR), and they are differentially expressed throughout human development. β -LCR is a distal control element that upregulates the expression of these structural genes. Fig. 5.1 shows the various globin genes located on chromosome 11.

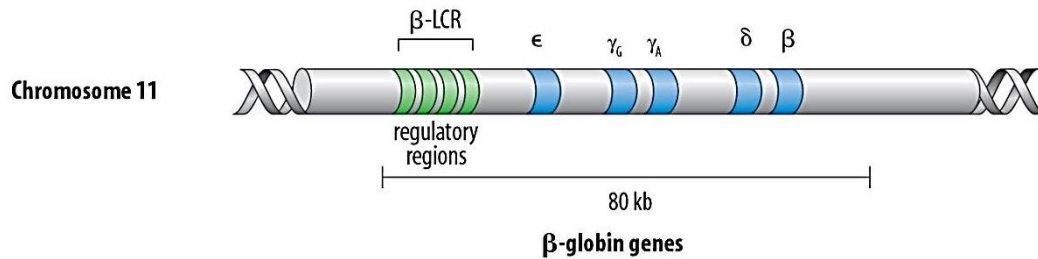


Fig. 5.1

- (a) Explain how β -LCR regulates the expression of structural genes.

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.....[3]

δ globin gene is expressed at a low level in adults. The low expression level of the gene is regulated by chromatin modification.

- (b) Describe one way by which chromatin modification can lead to low level of δ -globin gene expression.

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.....

.....[2]

The expression of globin genes in human can also be regulated at translational and post-translational level.

- (c)** Describe one way by which gene expression can be regulated at translational level.

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..... [2]

Haemoglobin and collagen are two of the most abundant proteins present in human.

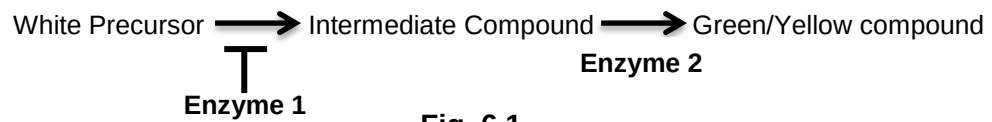
- (d)** Distinguish between the molecular structures of haemoglobin and collagen.

.....
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.....[2]

[Total: 9]

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6. Summer squash are a subset of squashes that are harvested when immature while the rind is still tender and edible. Nearly all summer squashes are varieties of *Cucurbita pepo*. The colour of squash fruits may be white, yellow or green. The colours are determined by two different genes located on separate chromosomes. Coloured pigments are synthesised enzymatically from a white precursor as shown in Fig. 6.1.



When pure-bred white plants cross with pure-bred green plants, the resulting F_1 offspring are all white. When the resulting F_1 offspring undergoes self-fertilisation, the results of F_2 generation are as follow:

White plants: 183

Green plants: 42

Yellow plants: 15

- (a) Construct a genetic diagram to explain the self-cross of F_1 generation and the results for the F_2 generation.

- (b) State the relationship between the two gene loci.

.....[1]

- (c) Explain how you could determine that a white coloured plant is heterozygous at both loci.

.....

[2]

In an investigation, the self-fertilisation of a white coloured plant and the results of the F_2 generation were analysed. A chi-squared analysis was carried out and the chi-squared value calculated was 2.10.

Table 6.1

D.O.F	Probability, p				
	0.1	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

- (d) Using the values in Table 6.1, draw appropriate conclusions as to whether the results of the cross actually followed the expected ratio you have predicted in (a).

.....

[3]

Fig. 6.2 shows the electron micrograph of a *C. pepo* cell undergoing a particular stage of meiosis.



Fig. 6.2

(e)

(i) Describe one event that occurs in Fig. 6.2.

.....
[1]

(ii) Explain how the second division of meiosis differs from mitosis.

.....

[2]

Table 6.2 shows the number of chromosomes and the mass of DNA in different nuclei. All nuclei come from the same cell.

(f) Complete Table 6.2.

Table 6.2

Nucleus	Number of chromosomes	Mass of DNA / arbitrary units
At prophase of mitosis	40	60
At telophase of mitosis		
In a gamete cell		

[2]

[Total: 15]

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7. Bats represent the only known example of flight in mammals. They are the only animals with sophisticated laryngeal echolocation that is where ultrasonic calls are produced in the larynx. Many species of bats hunt flying insects at night. They are able to use echolocation (sound waves) to help them find their prey in the dark.

The oldest known fossilised remains of bats (50-60 million years ago) show that ancestral bats looked very much like they do today. However, fossil evidence is inconclusive whether or not the oldest species known to date (*Onychonycteris finneyi*) was capable of echolocation.

- (a) Explain how the ability to use laryngeal echolocation may have evolved from an ancestral species that cannot use echolocation.

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..... [3]

The geographical distribution of *Pipistrelles Pipistrellus* (common pipistrelle) extends across most of Europe, North Africa, and south-western Asia. It is one of the most common bat species in the British Isles. Originally, it was thought that all pipistrelle belong to the same species, *Pipistrelles Pipistrellus*. In 1999, the common pipistrelle was split into two species on the basis of different-frequency echolocation calls. The common pipistrelle uses a call of 45 kHz, while the *Pipistrellus Pygmaeus* (soprano pipistrelle) echolocate at 55 kHz.

Since the two species were distinguished, a number of other differences, in appearance, habitat and food, have also been discovered. It has been postulated that the distinct phenotypic differences between *Pipistrellus Pipistrellus* and *Pipistrellus Pygmaeus* may have arisen in the absence of geographical barriers.

- (b)** Explain why molecular homology is better than anatomical homology in determining evolutionary relationship between the two species.

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.....[2]

- (c)** Describe one way by speciation may have arisen in the absence of geographical barriers.

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.....[2]

Differences in the *cytochrome b* DNA sequence of several bat species from different regions of Europe were measured and plotted against time since divergence from the primitive ancestor (MYA) as seen in Fig. 7.1.

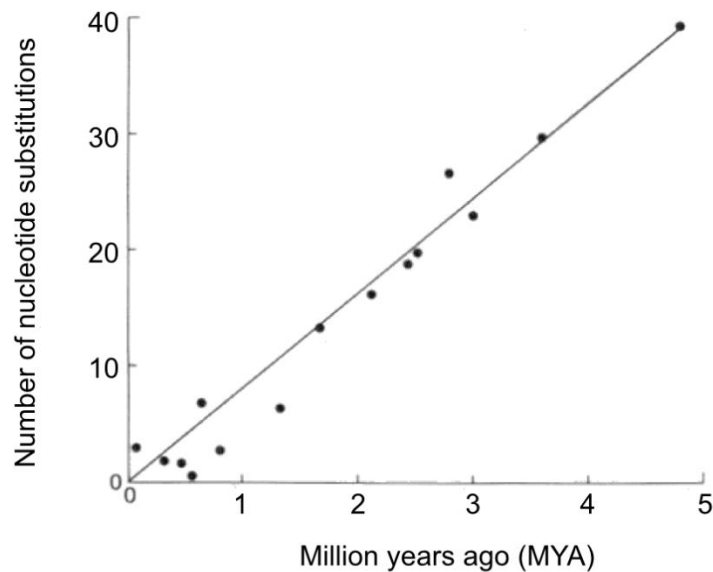


Fig. 7.1

- (d) Describe how these differences support the neutral theory of molecular evolution.

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.....[3]

[Total: 10]

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8. Fig. 8.1 shows a post-synaptic neurone, **Z**, forming synapses with adjacent neurones including Neuron **A** and **B**. The arrow in Fig. 8.1 shows the direction of the nerve impulse.

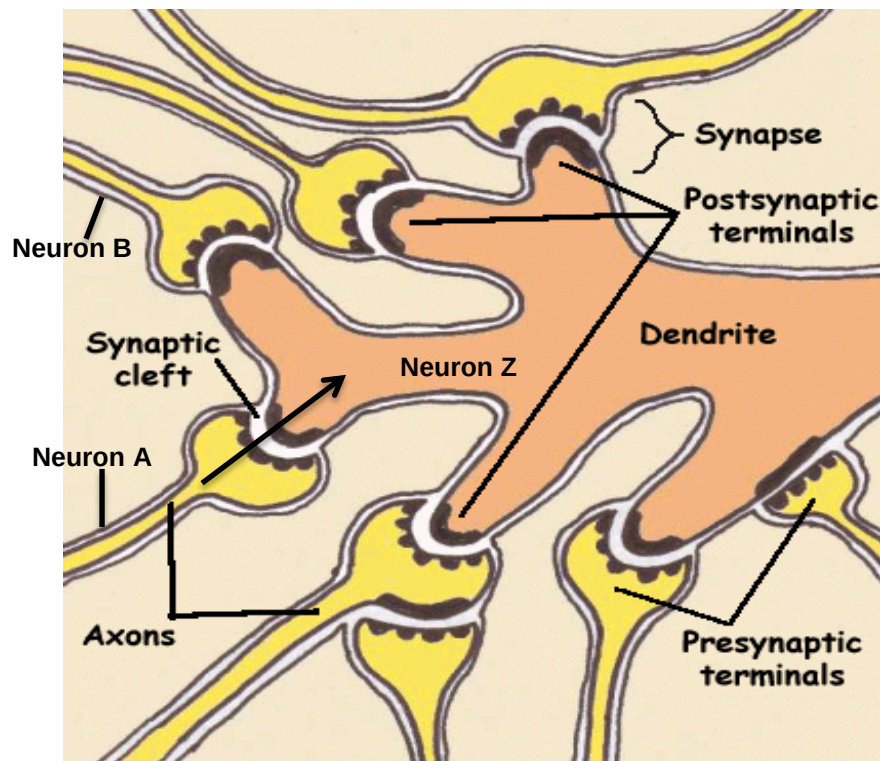


Fig. 8.1

- (a) With reference to Fig. 8.1, describe how transmission of nerve impulse across the synapse is believed to have taken place.

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.....[3]

When brain experiences an abundance of stress, it can be caused by a surplus of adrenaline. To neutralize this extra adrenaline, the brain produces neurotransmitters, one of which is GABA, that have inhibitory effects upon the nervous system.

Neurone **A** releases adrenaline; a neurotransmitter which stimulates the production of nerve impulses in neurone **Z**. Neurone **B** releases GABA, which inhibits the production of nerve impulses in Neurone **Z**.

- (b) Explain how Neurone **B** inhibits the production of nerve impulses in neurone **Z**.

.....
.....
.....[2]

- (c) Describe one mechanism that helps to restore the resting membrane potential of pre-synaptic neuron **A**.

.....
.....
.....[2]

[Total: 7]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

9.

- (a)** Describe the structure of phospholipids and explain how the structure enables them to form the cell surface membrane. [4]
- (b)** Describe the role of nicotinamide adenine dinucleotide in aerobic and anaerobic respiration. [7]
- (c)** Describe the structure of the deoxyribonucleic acid and explain how they are suited for replication. [9]

[Total: 20]

10.

- (a)** With reference to the key stages of cell signalling, describe how a liver cell respond to insulin stimulation. [7]
- (b)** Outline the main stages of Calvin cycle. [8]
- (c)** Explain the relationship between the frequency of sickle cell anaemia allele and the number of malaria cases. [5]

[Total: 20]

End of Paper